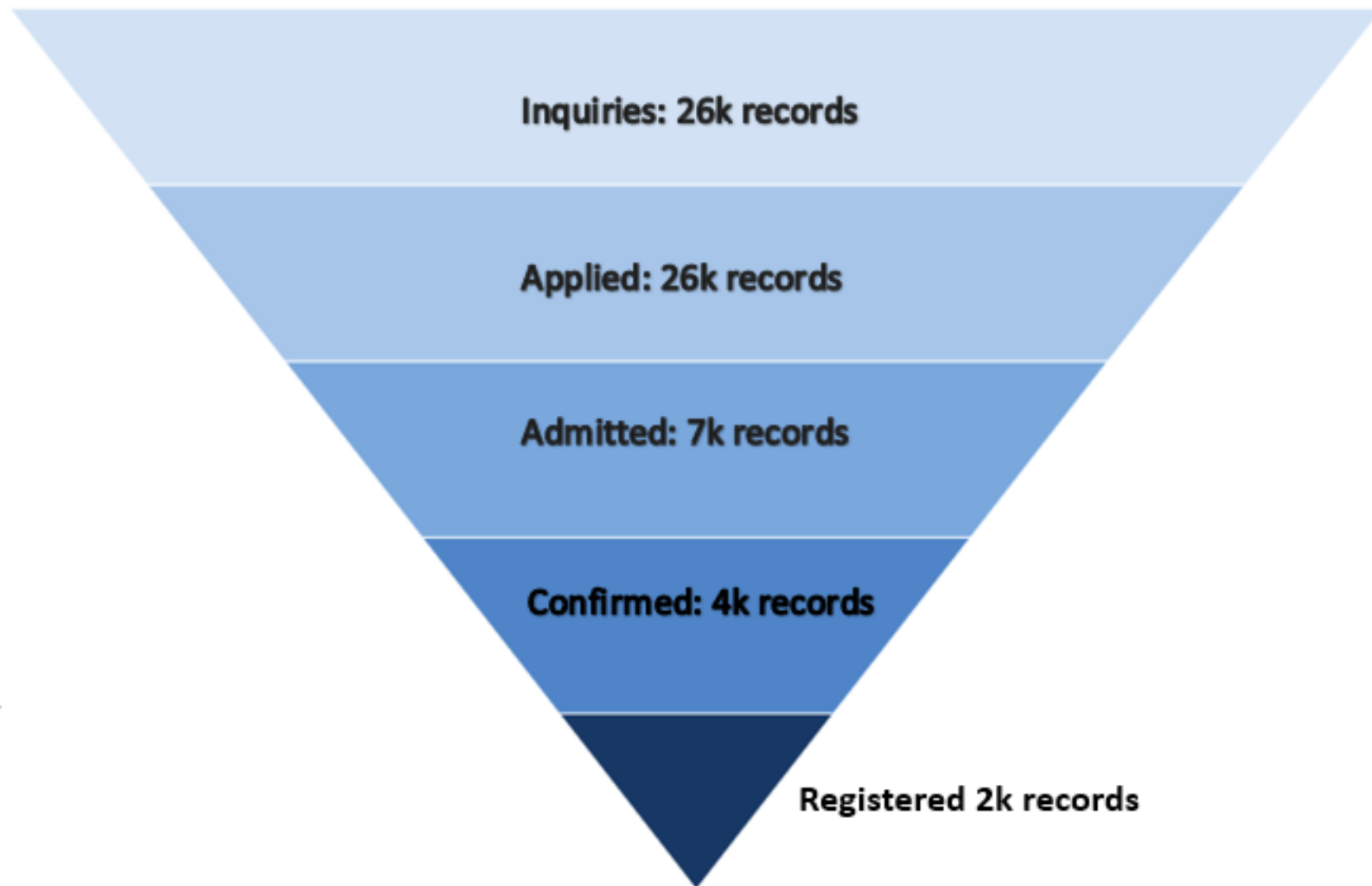




Graduate Admissions Insights

By: Sandeep, Jayathi, Patra, Thao, and Uyen

From Interest to Enrollment: The Funnel



How we approached it



Define funnel stage



Clean data

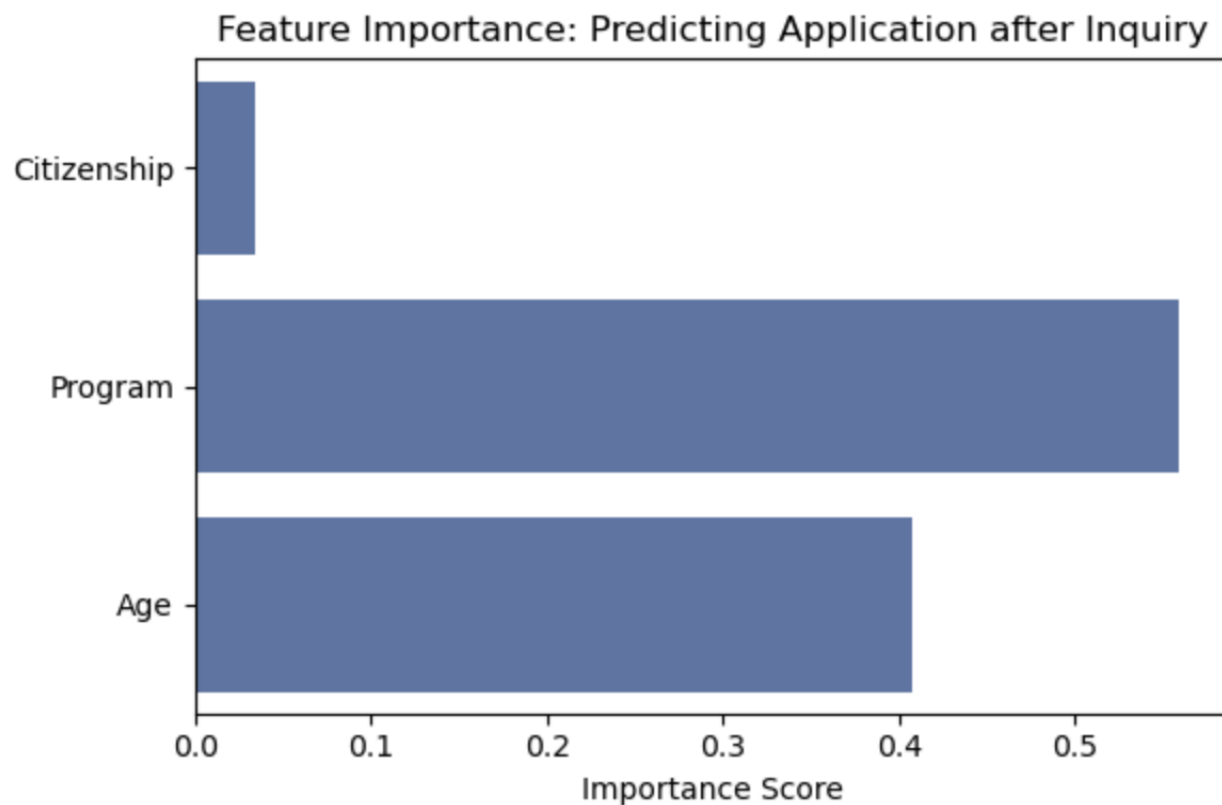


Apply statistical tests / predictive models



Generated insights

Inquiries / Stealth Applicants

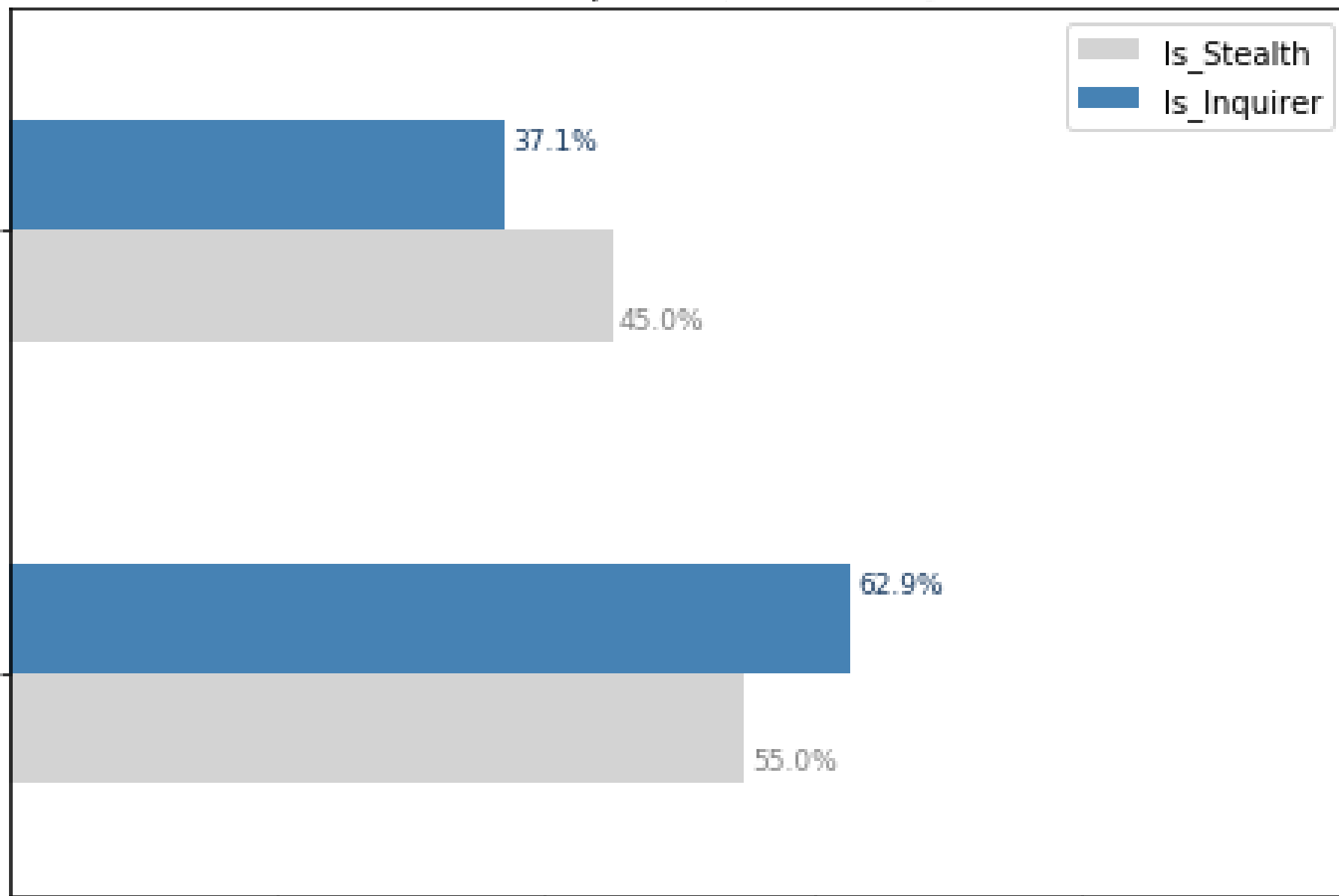
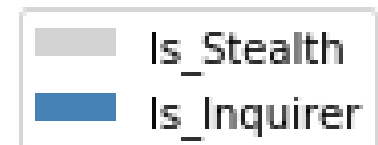


- **Top Programs**
- **Citizenship:**
- **Age**

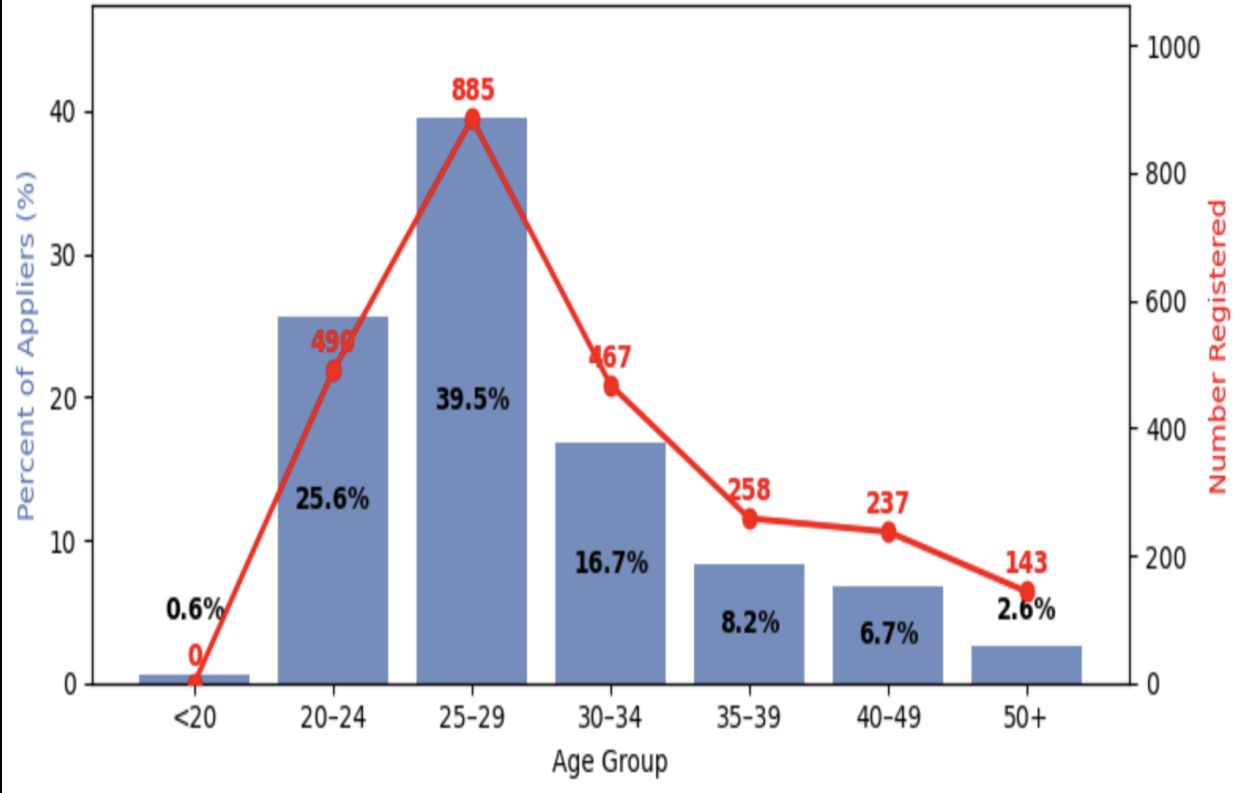
Citizenship Group

Foreign National

US/PR

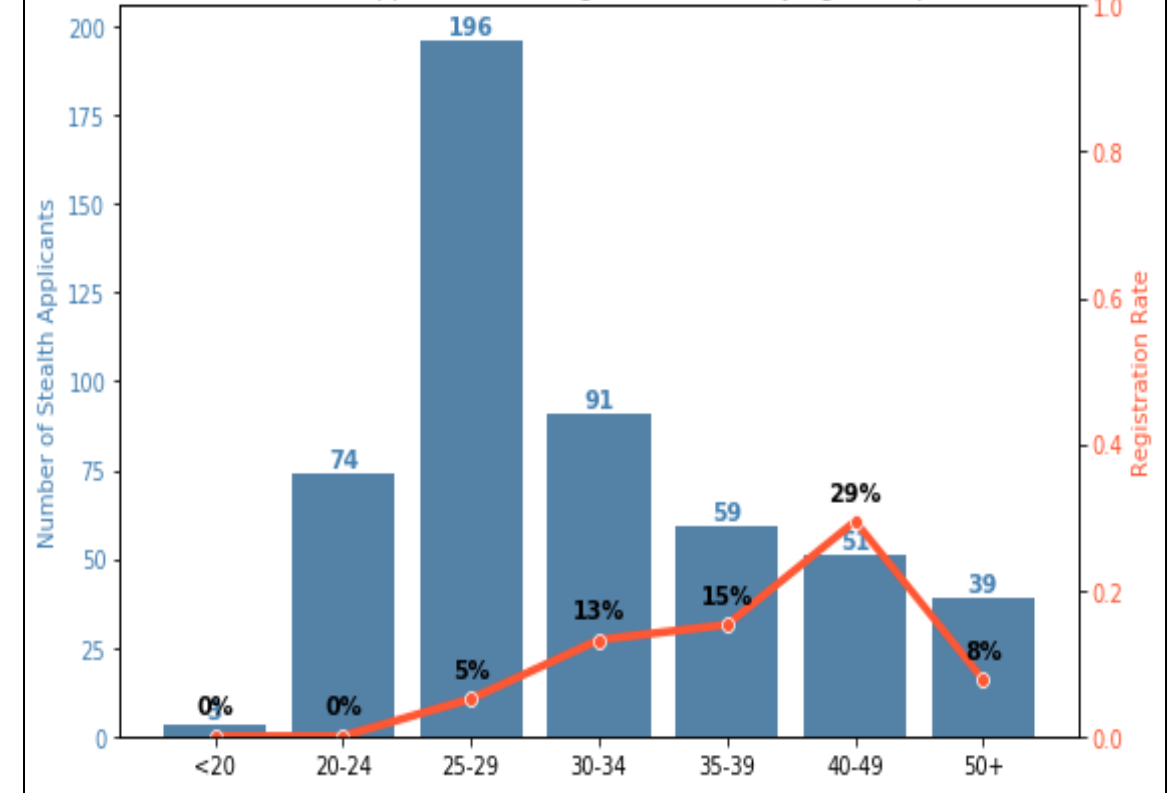


Appliers and Registration Rate by Age Group

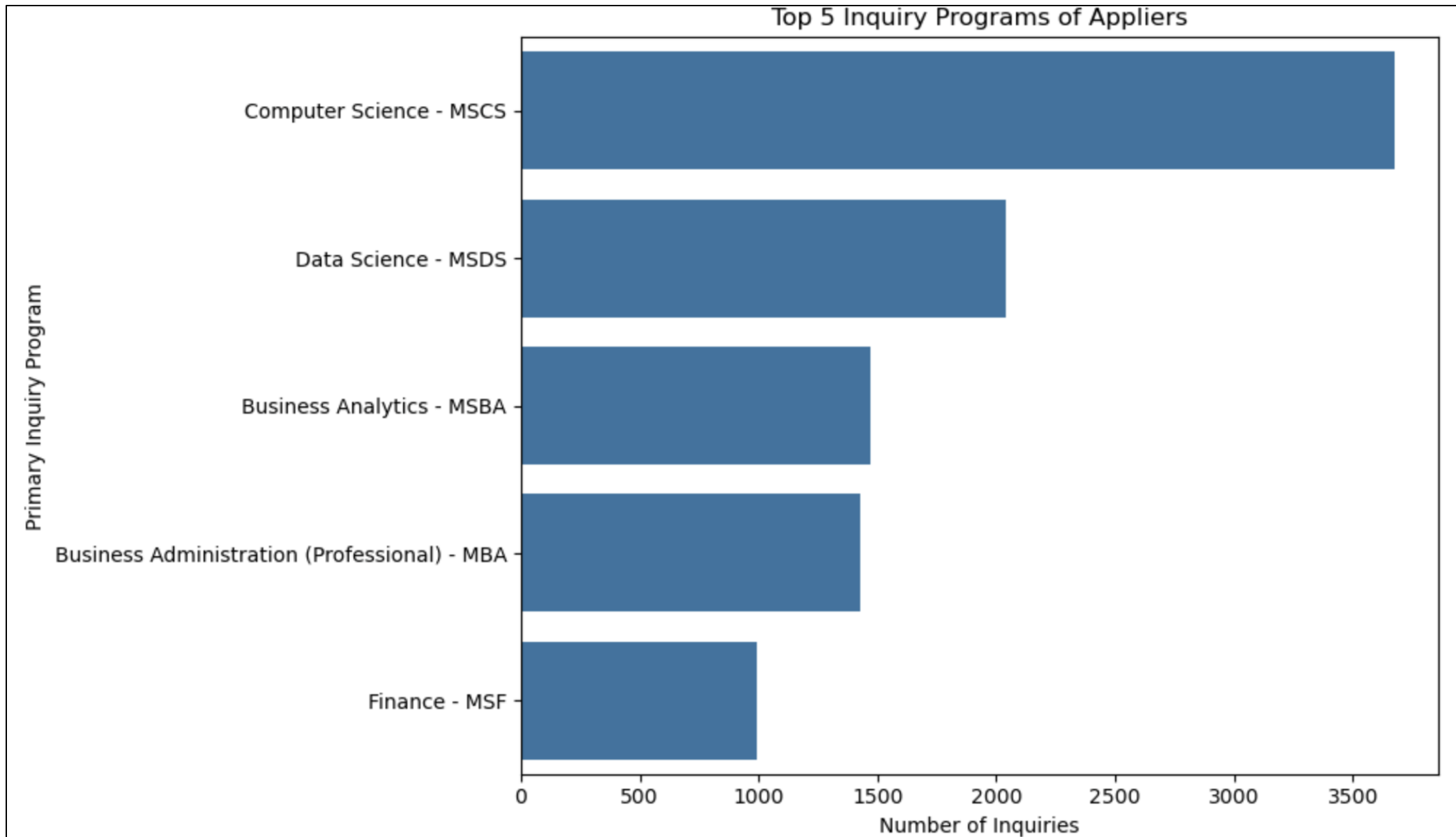


Younger Applicants
Inquire and Actually Register

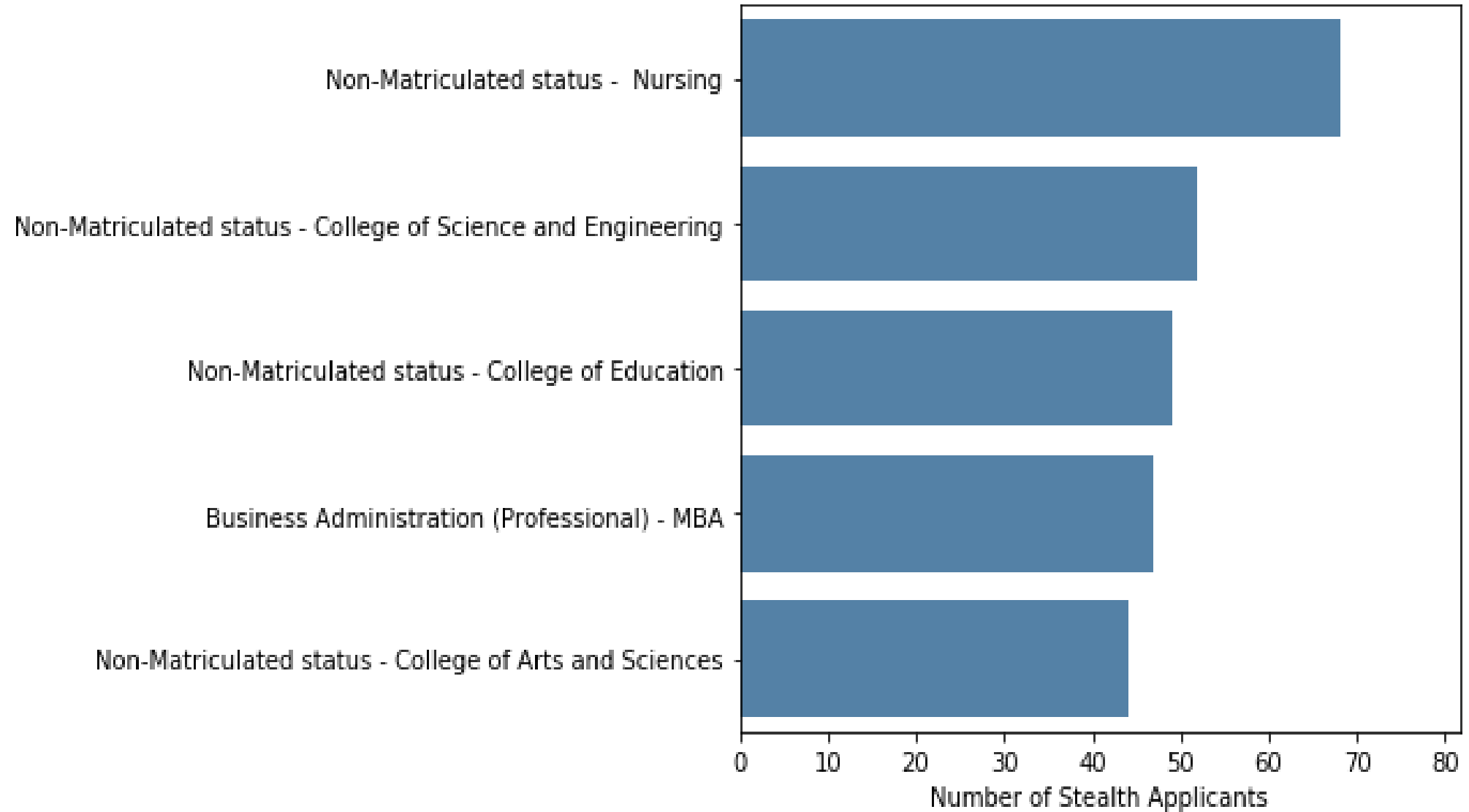
Stealth Applicants and Registration Rate by Age Group

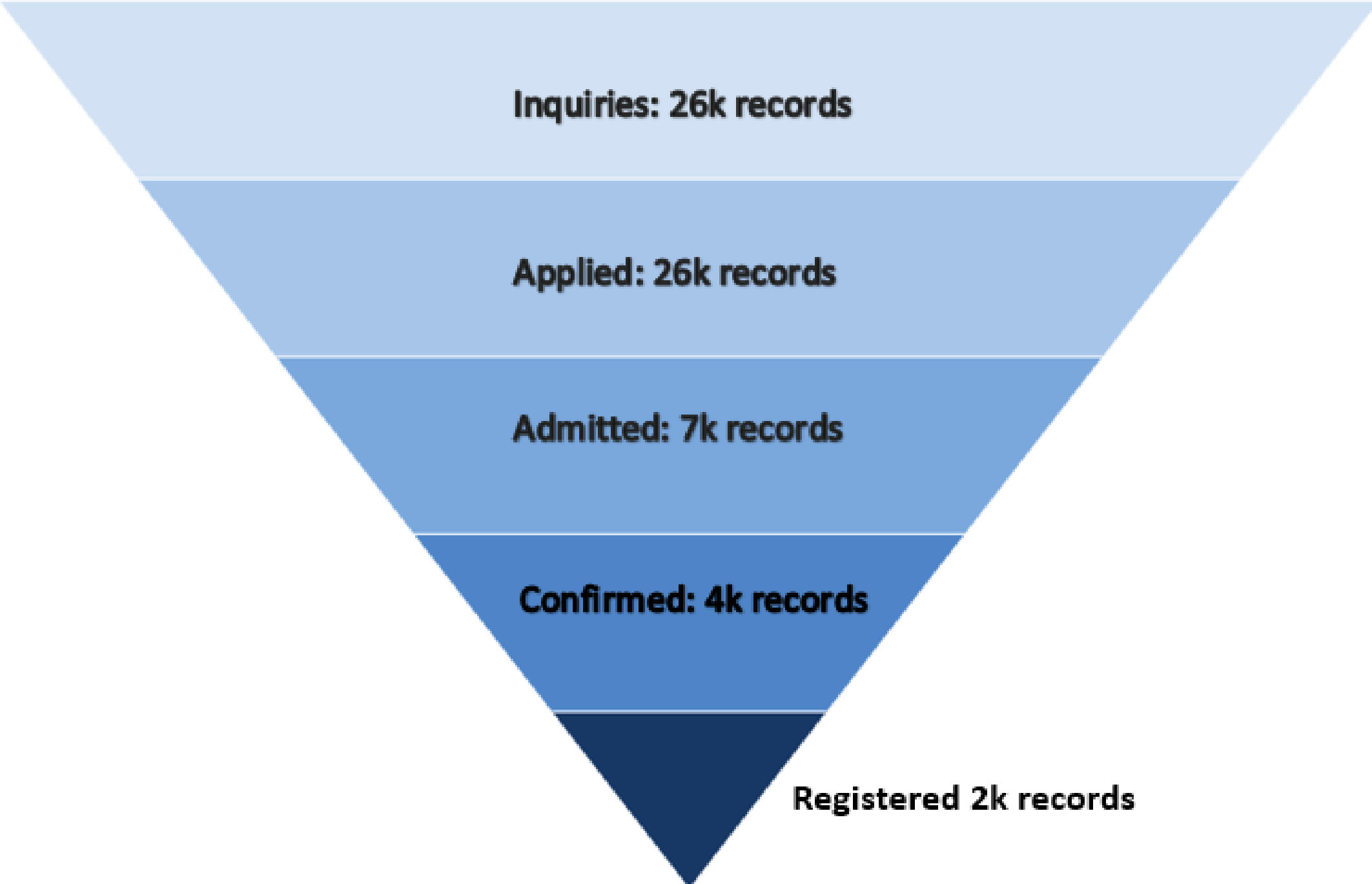


Older Stealth Applicants
Registered (Younger Applicants Do Not)



Top 5 Programs by Number of Stealth Applicants





Inquiries: 26k records

Applied: 26k records

Admitted: 7k records

Confirmed: 4k records

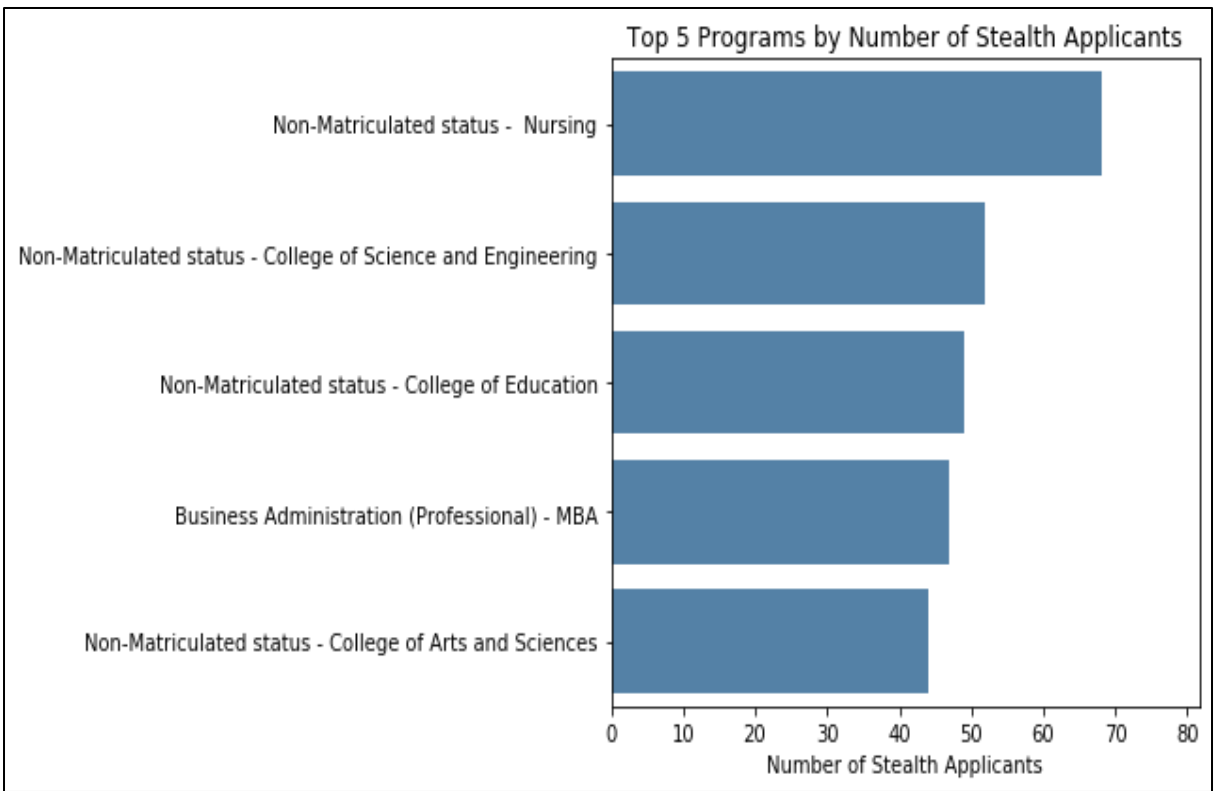
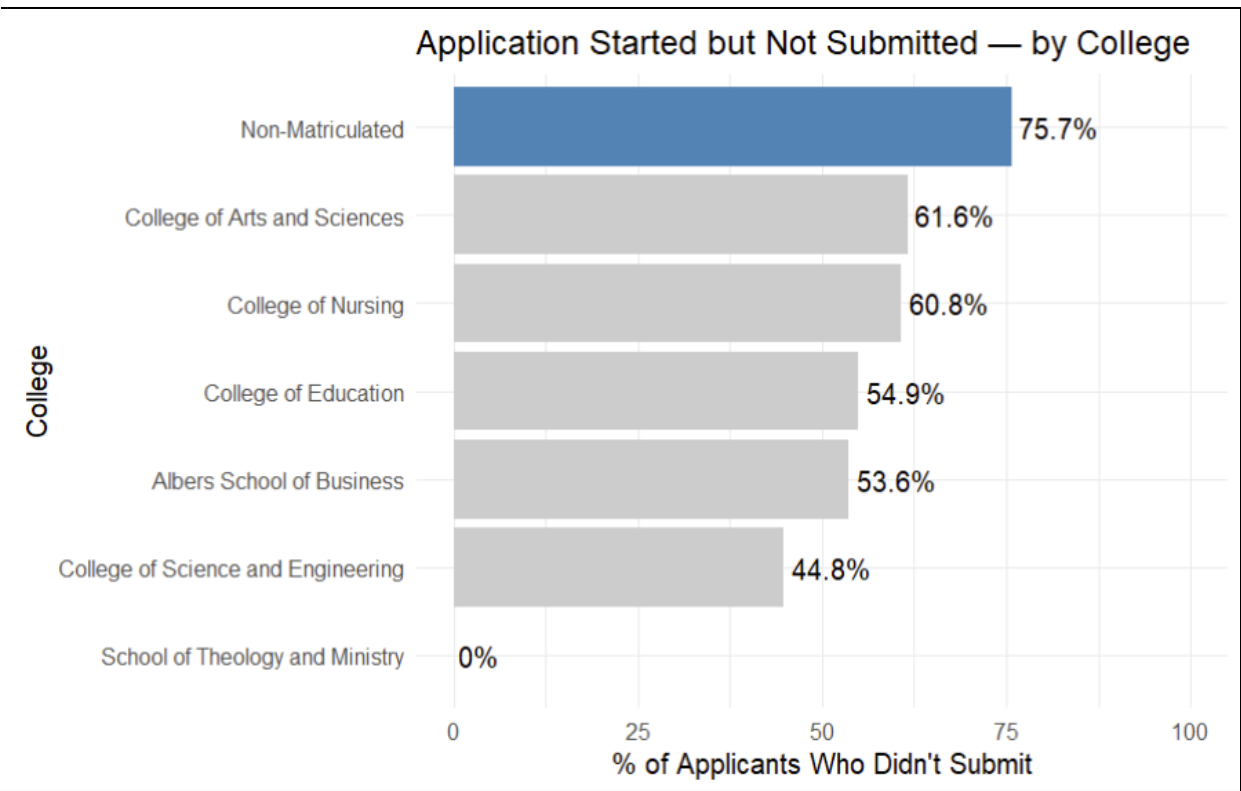
Registered 2k records

Drop off applicants

1. Who Starts But Doesn't Submit?

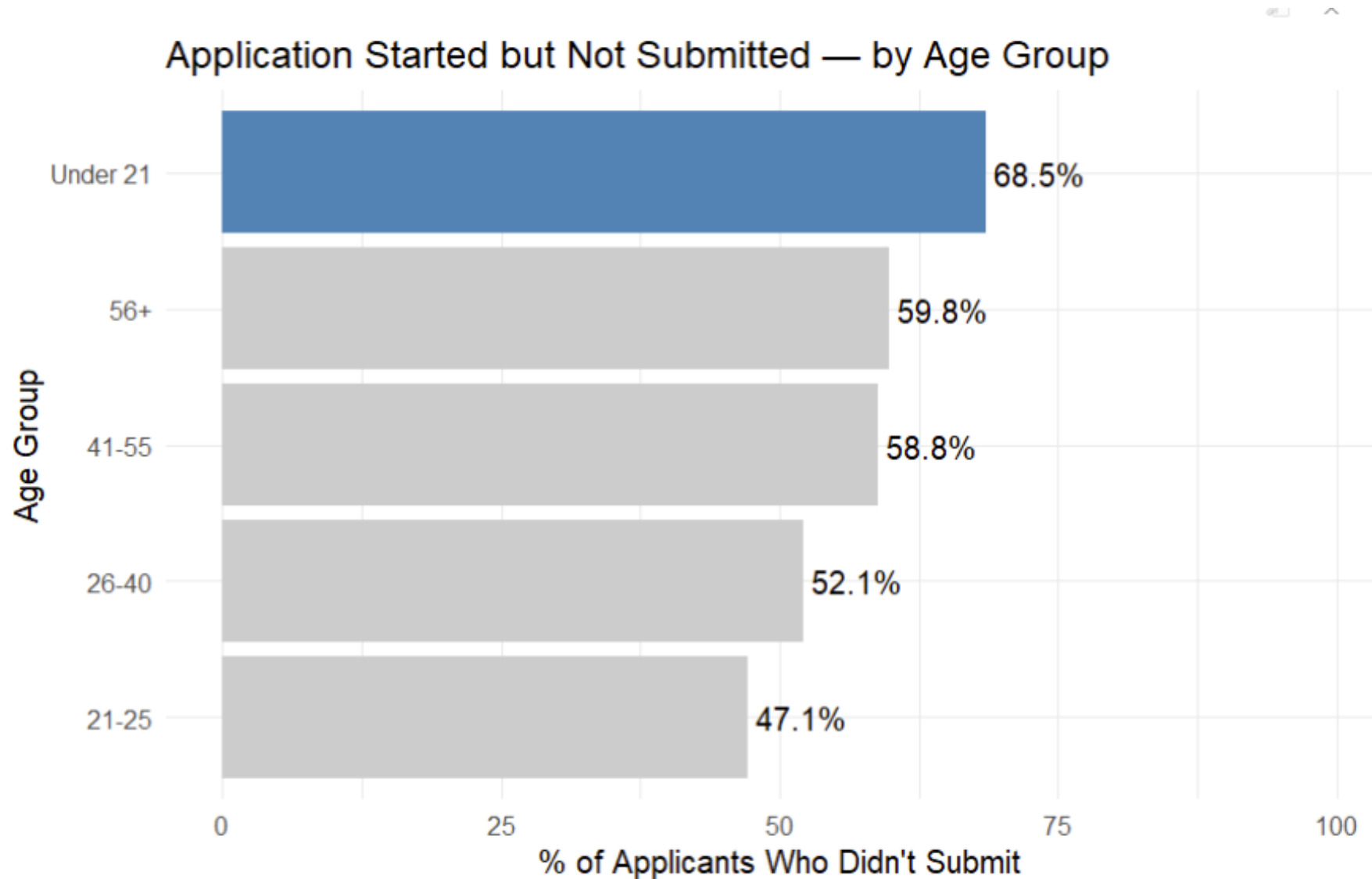
- **Key Factors:**
 - College
 - Age Group
 - Race Group

Who Starts But Doesn't Submit? - By College

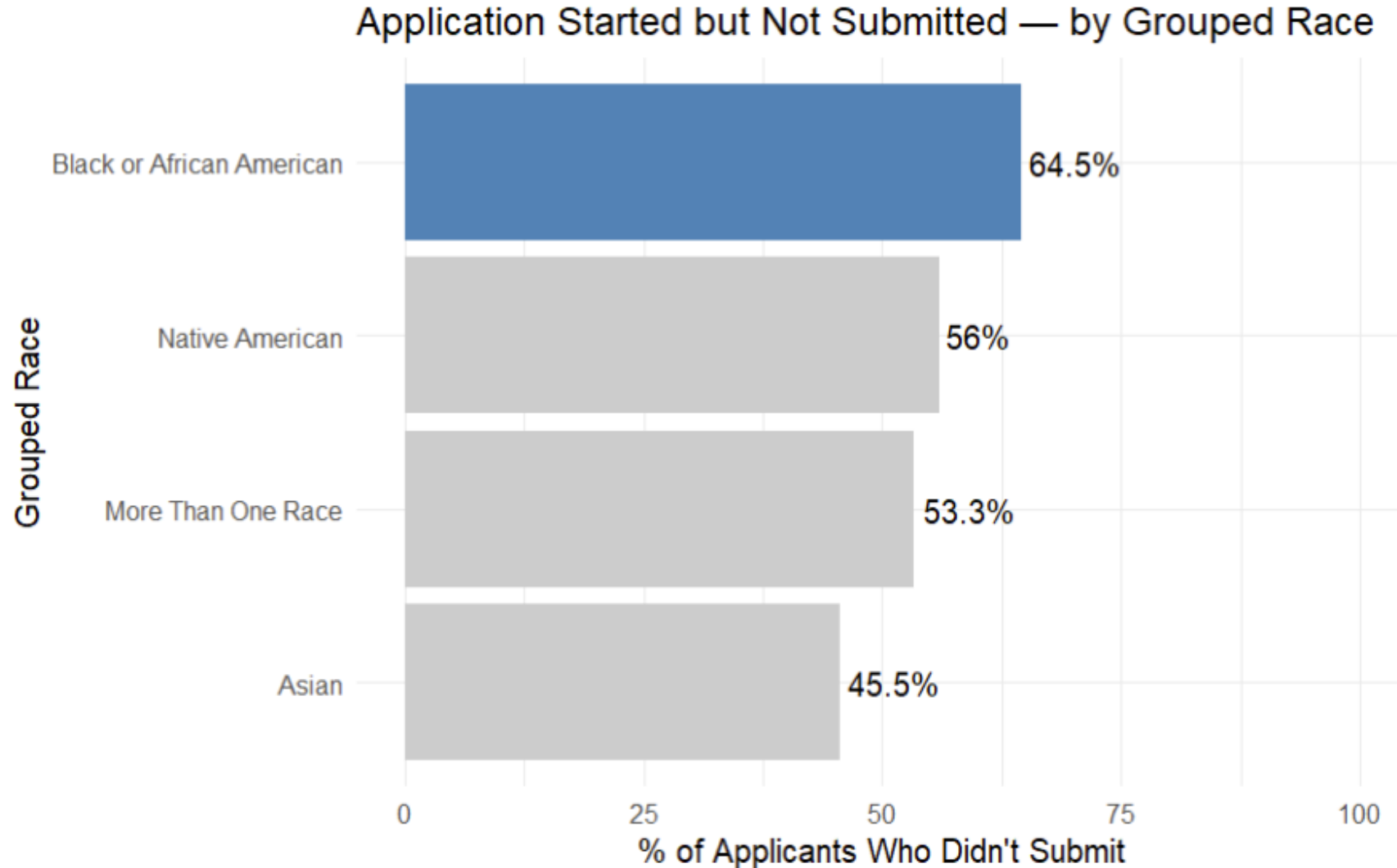


Stealth applicants in non-matriculated programs are the most likely to start an application but not follow through

Who Starts But Doesn't Submit? - By Age



Who Starts But Doesn't Submit? - By Race

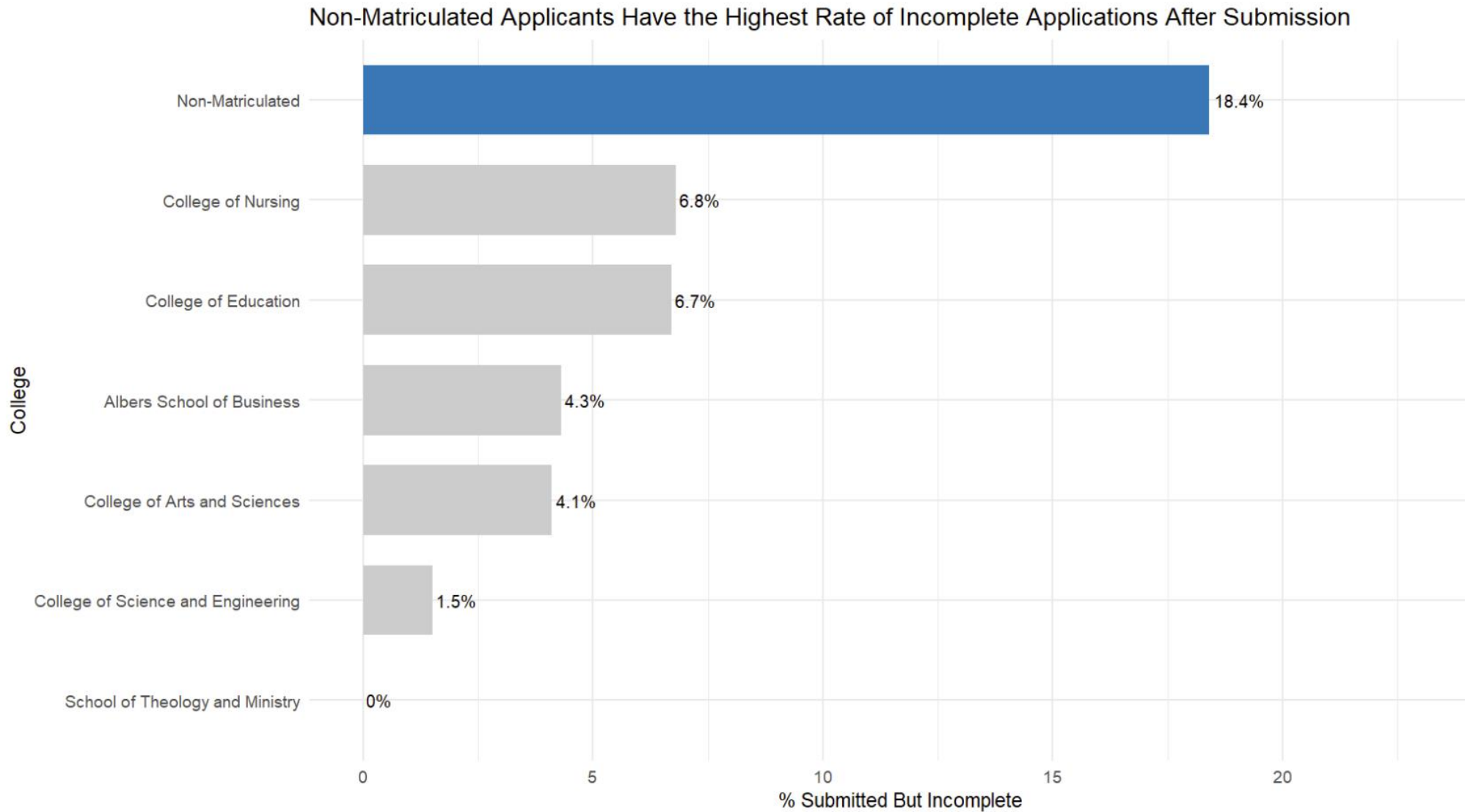


Drop off applicants

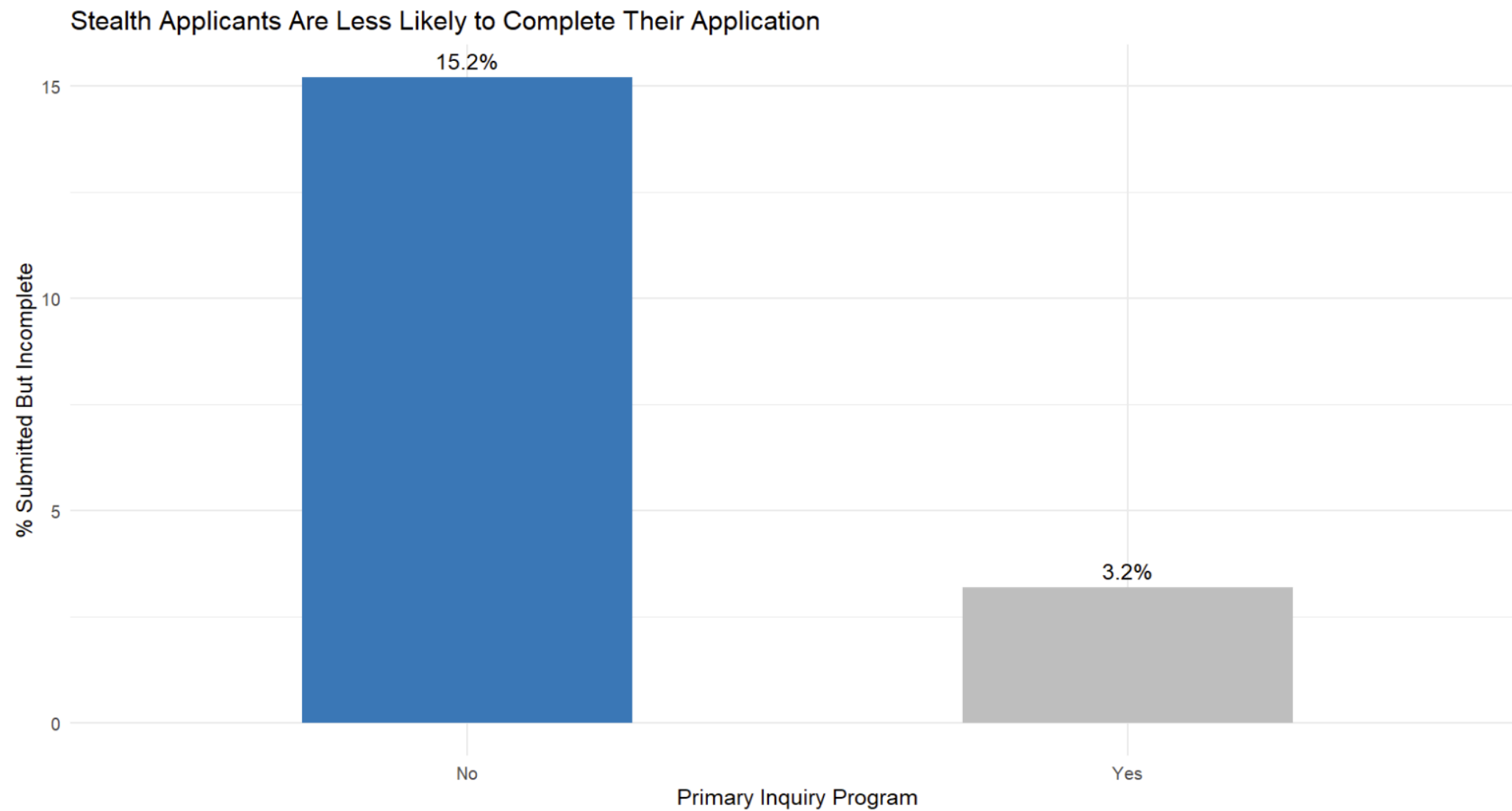
2. Who Submit But Doesn't Complete?

- **Key Factors:**
 - College
 - Primary Inquiry
 - Application Term

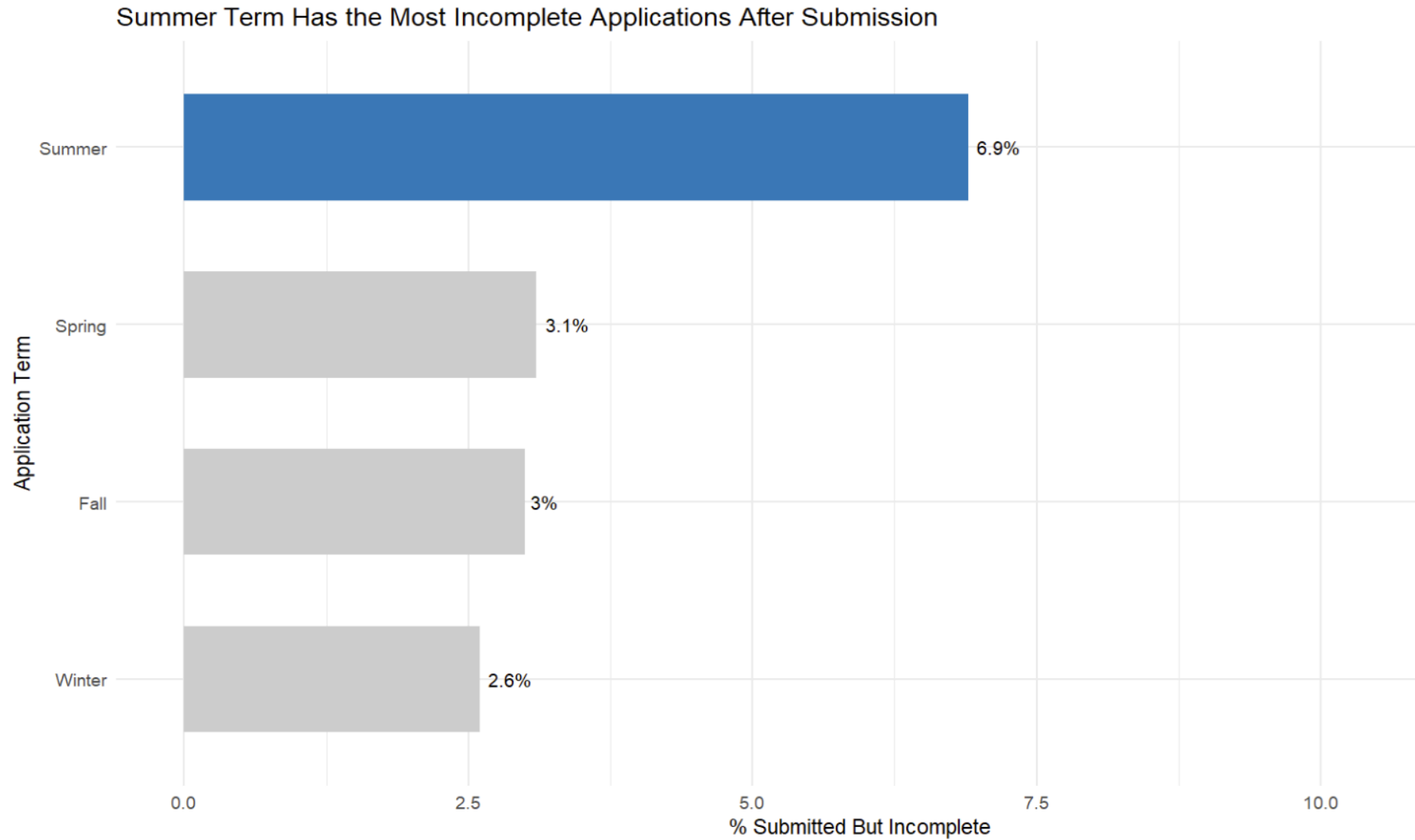
Who Submit But Doesn't Complete? - By College



Who Submit But Doesn't Complete? - By Inquiry



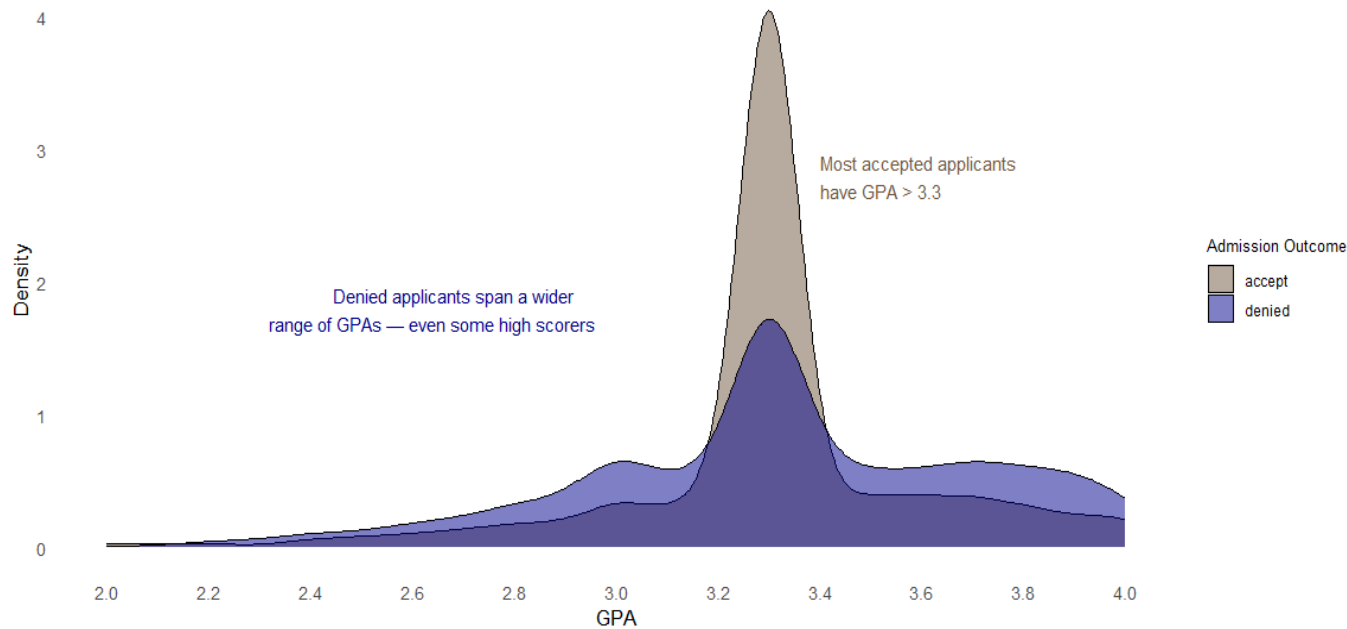
Who Submit But Doesn't Complete? - By Application term



What Factors Are Associated with Graduate Denial?

GPA Distribution by Admission Outcome

Many denied applicants have lower GPAs, but others are rejected despite strong academic performance

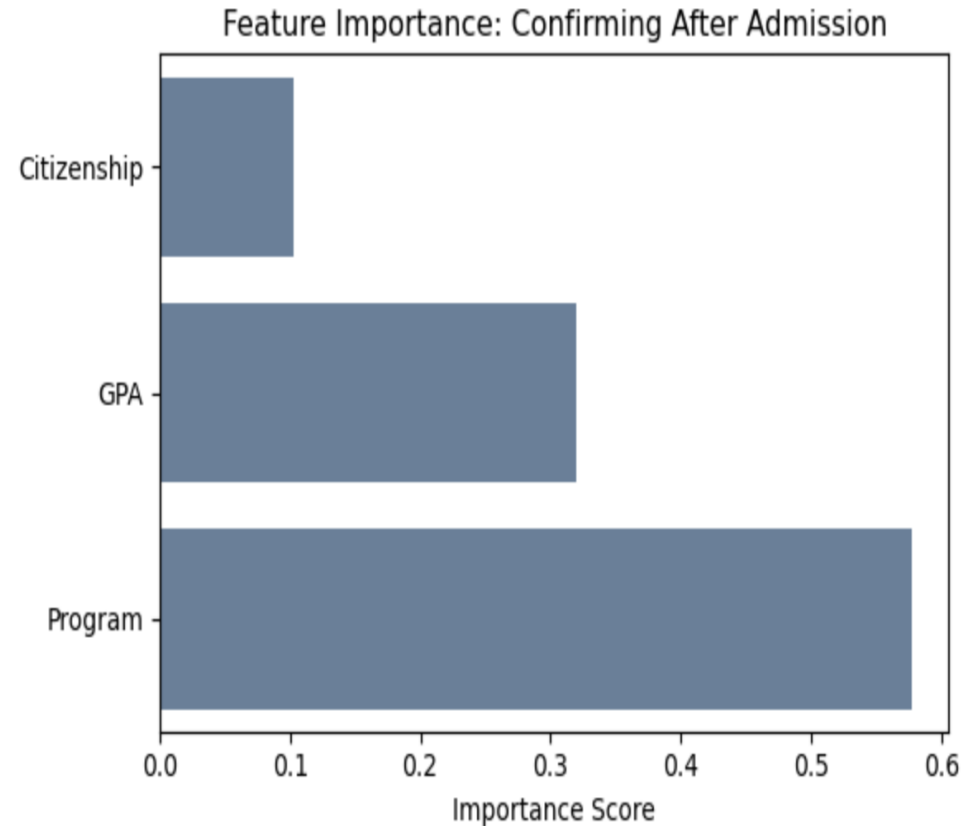


“Applicants Aged 45+ Have the Highest Denial Rate Across Age Groups” – 23.5%

“Domestic Applicants Are Denied More Often Than International Students” - 29.2%

Understanding Applicants who get admitted but don't confirm

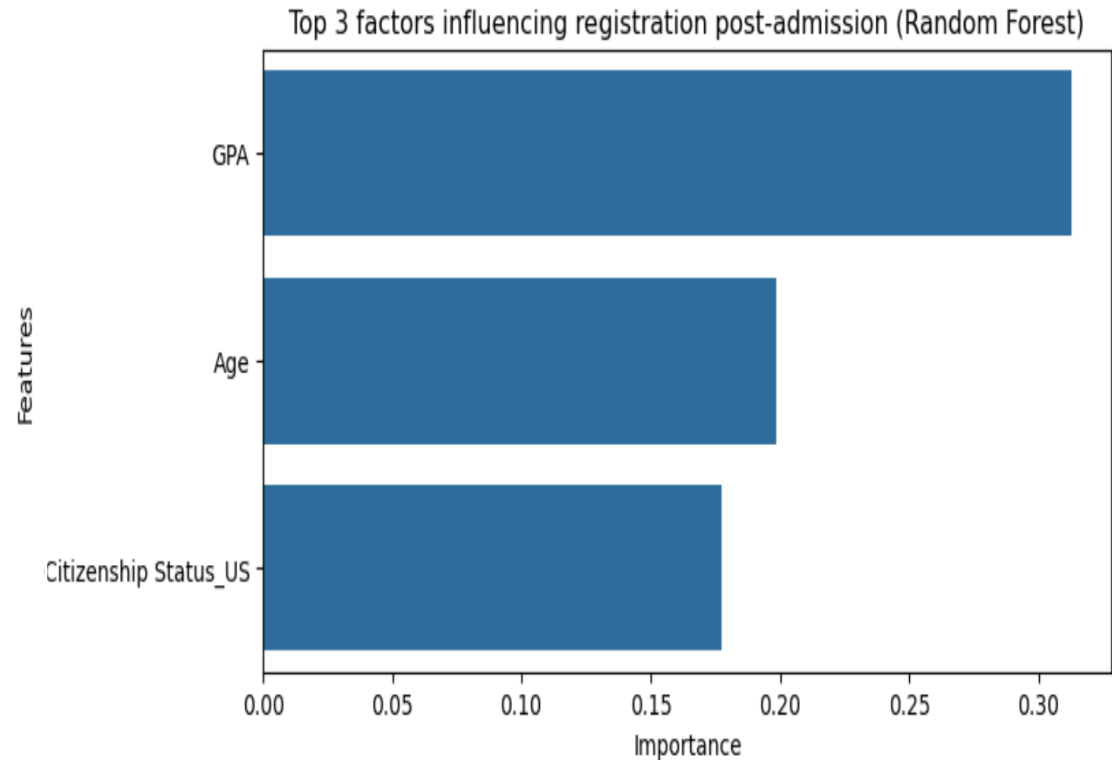
- Higher non-confirmation rates among **international students**.
- Non-confirmation spikes for **GPA**s below 3.2.
- Programs like **MSCS**, **MSDS**, and **MSBA** lead in non-confirmations.
- **Younger applicants** (early 20s) show higher non-confirmation rates.



[Visuals Link](#)

Who Doesn't Register After Confirming?

- **GPA:** 72.7% of applicants with GPAs between 3.0 and 4.0 did not register.
- **Age:** 66.3% of non-registrants are aged 22–30.
- **Citizenship:** 83.7% of non-registrants are foreign nationals.

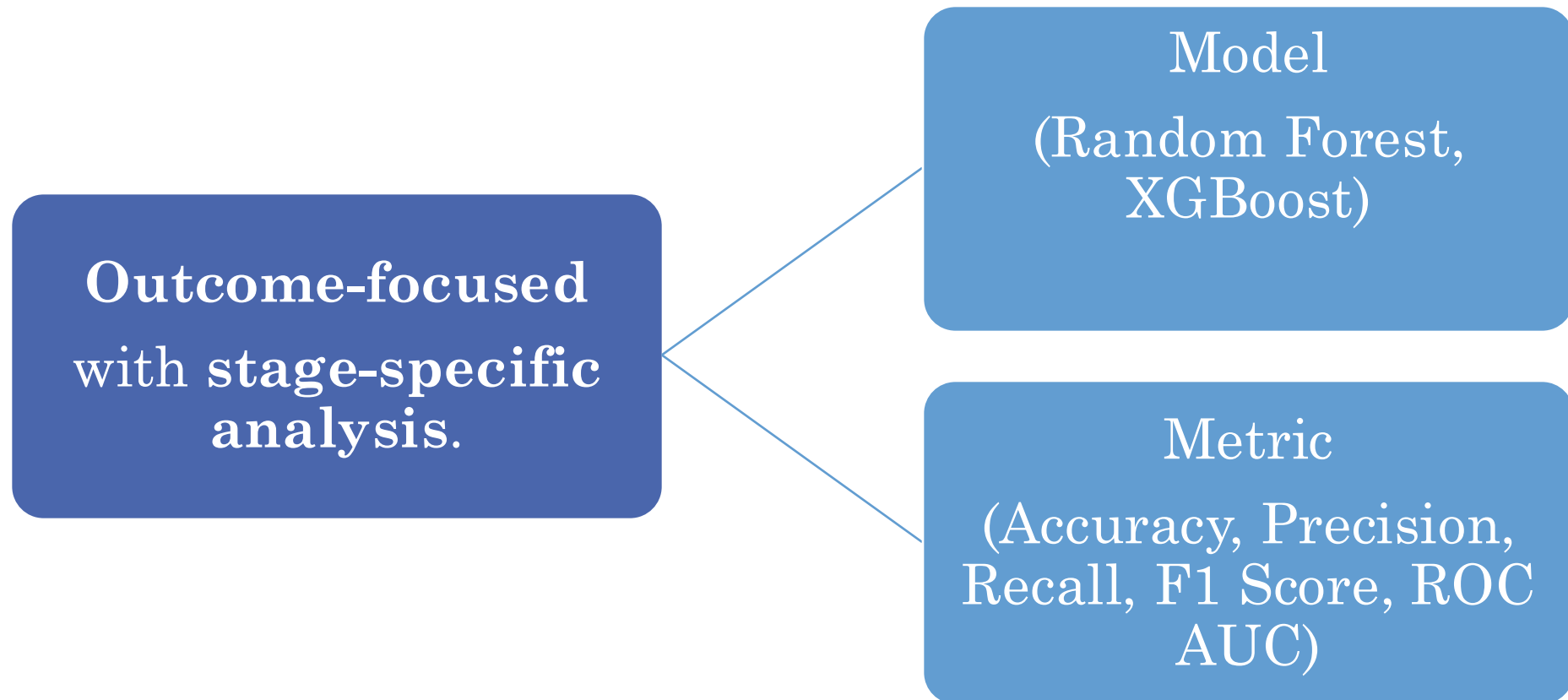


[Visuals link](#)

Recommendations

- Provide personalized support and guidance at every stage, from inquiry to registration.
- Actively reach out to applicants who do not confirm or register to understand their challenges and offer tailored solutions.
- Assist with exploring alternate programs or deferral options to keep applicants engaged.
- Enhance communication about scholarship opportunities and application guidance to address financial concerns.
- Focus on proactive engagement to ensure applicants feel supported throughout the enrollment journey

Hypothesis: There is identifiable data that can predict someone enrolling in graduate school at SU → **Yes**



Technical Appendix

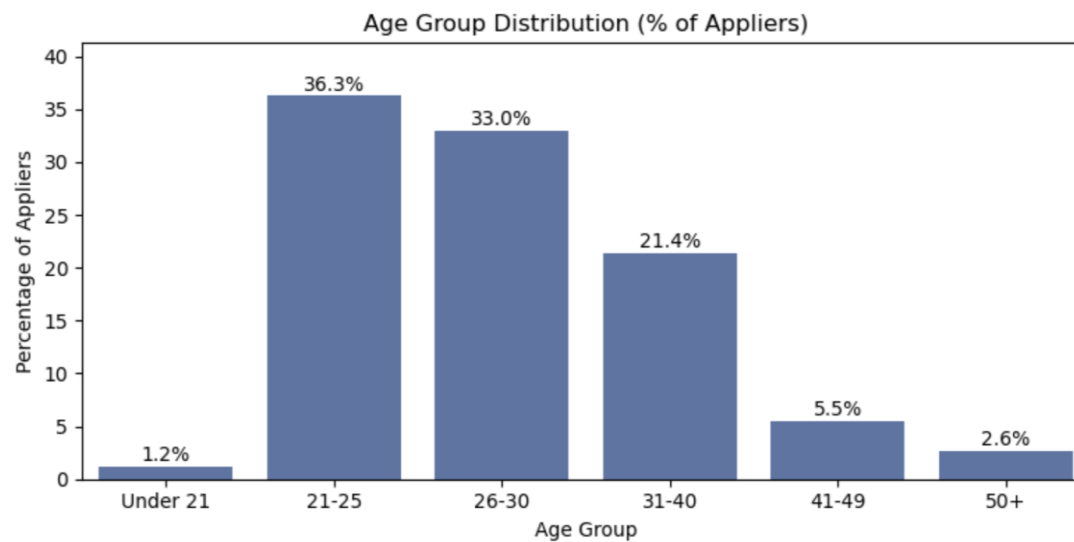
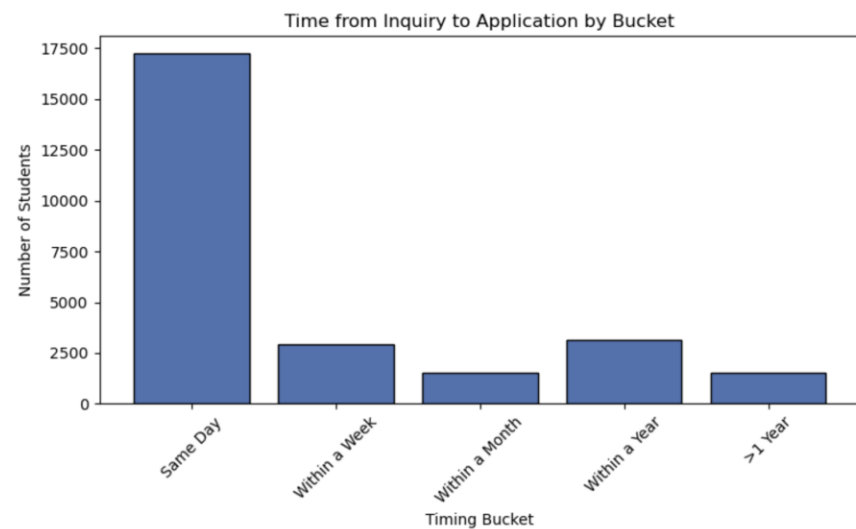
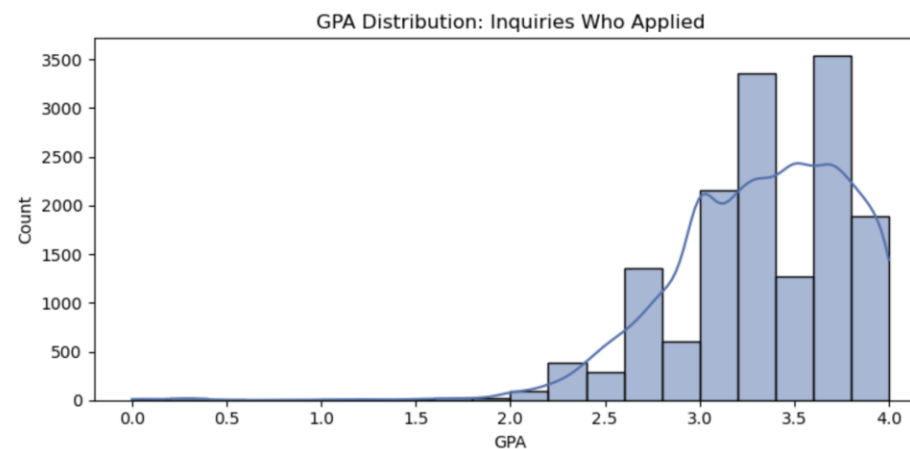
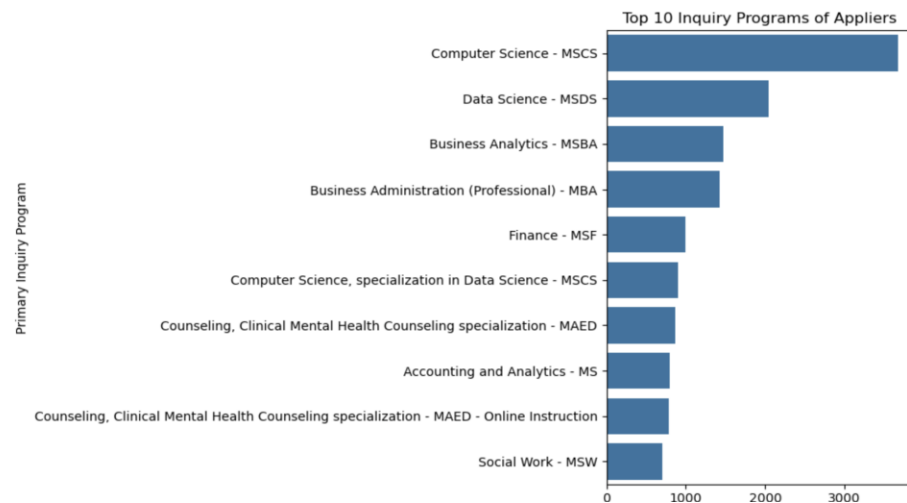


Models/Metrics

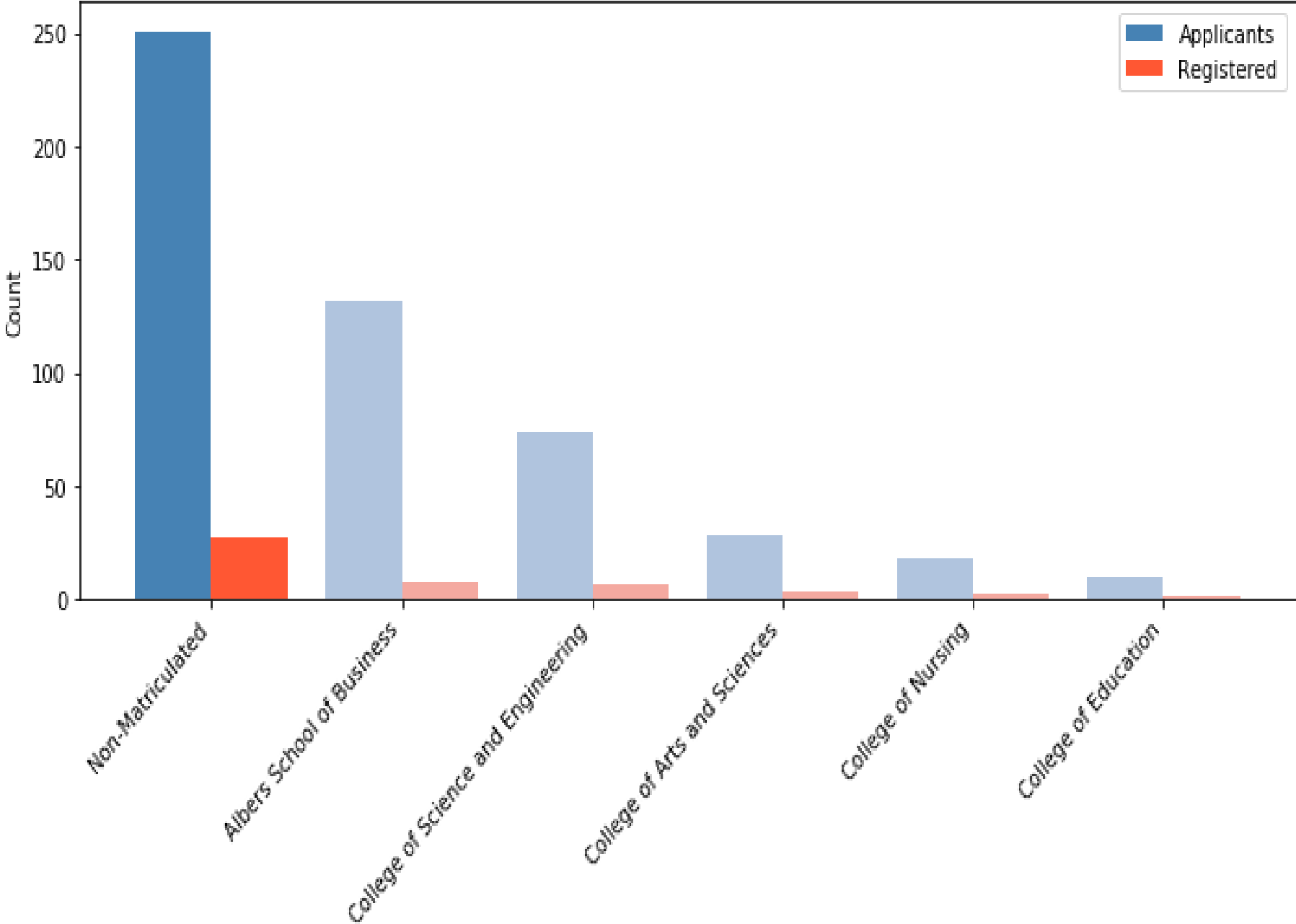
['College', 'Primary Inquiry Program', 'Application Program', 'Sex', 'Citizenship Status', 'Age Group', 'Application Term', 'Grouped Race', 'Scholarship Amount', 'Received Scholarship', 'Student Classification', 'GPA Group']

Model	Accuracy	Precision (1)	Recall (1)	F1 Score (1)	ROC AUC
Random Forest	81.5%	28.8%	41.7%	34.1%	0.7616
RF + SMOTE	78.6%	27.9%	54.9%	37.1%	0.7736
XGBoost	81.6%	29.9%	68.1%	41.6%	0.8571
Logistic + SMOTE	65.9%	23.0%	83.9%	36.1%	0.8006

- XGBoost best balance — high accuracy, recall, F1, and ROC AUC.
- Logistic + SMOTE maximum recall but sacrifices precision and accuracy.
- Random Forest has high accuracy but misses many enrollees.
- RF + SMOTE improves recall over baseline but still not as strong as XGBoost.



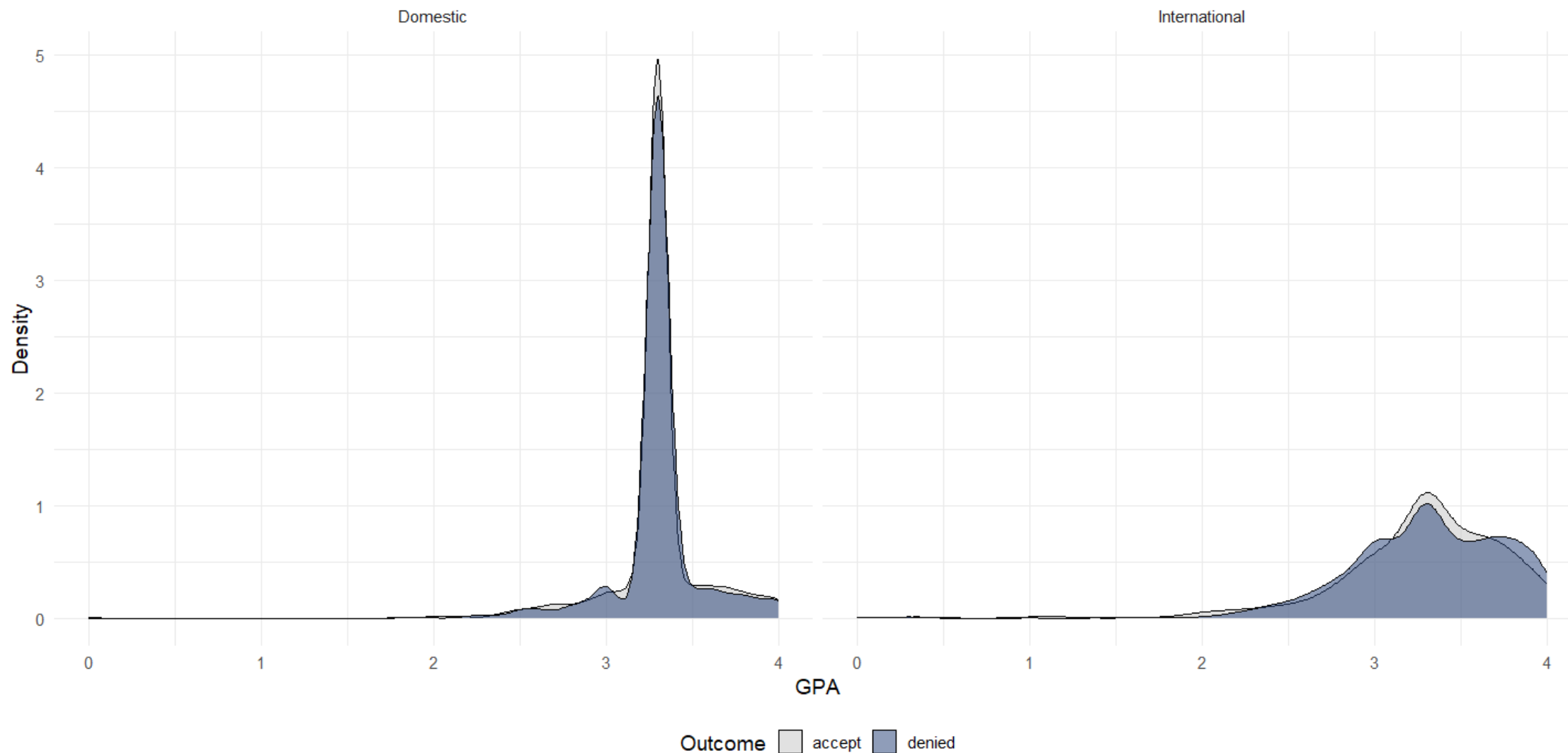
Stealth Applicants vs Registered by College (Non-Matriculated Highlighted)



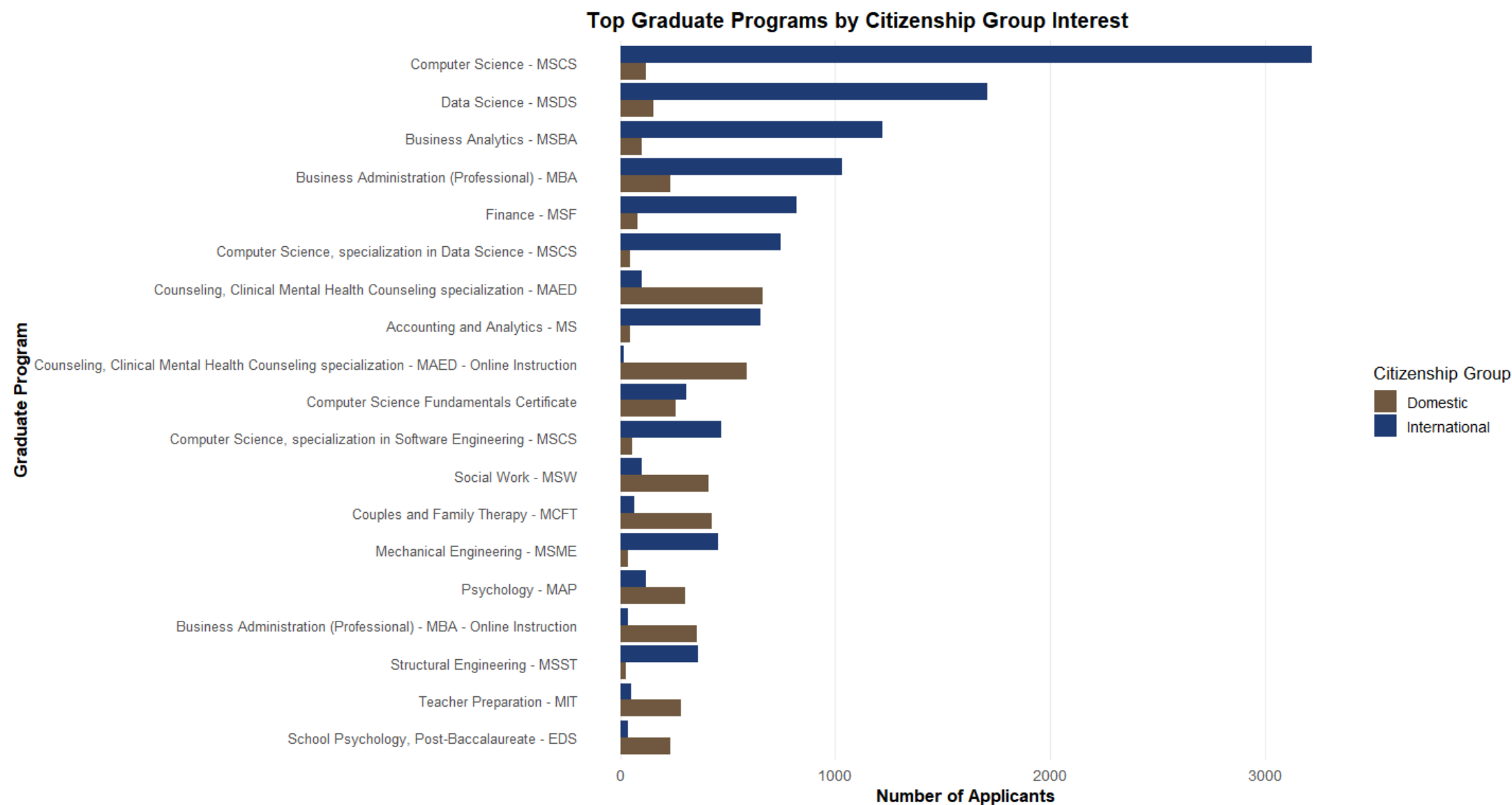
Why might domestic applicants be denied more?

GPA Distribution by Citizenship Group and Admission Outcome

Even at similar GPA levels, domestic applicants face higher denial rates

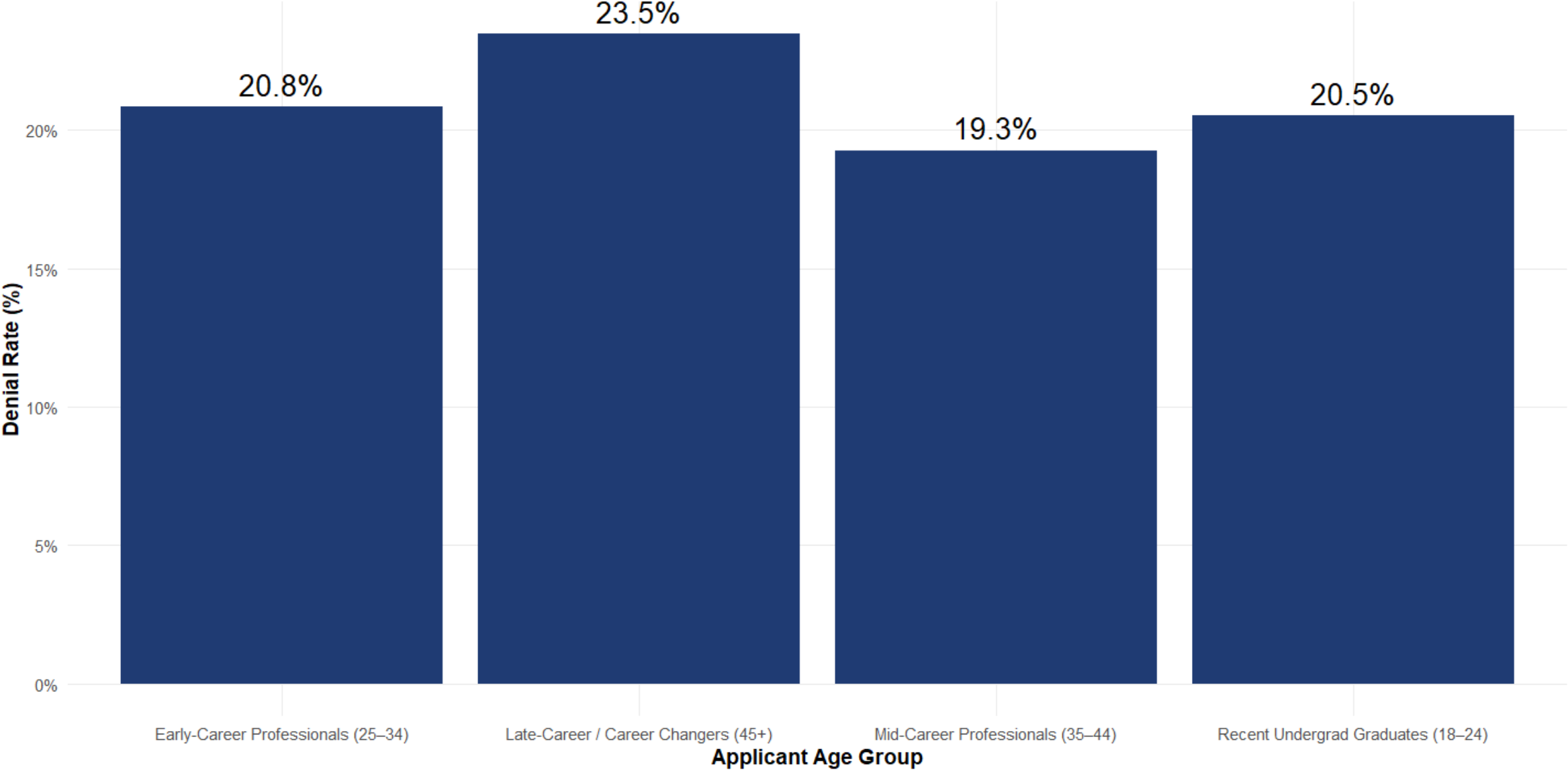


Domestic and International Applicants Show Different Program Preferences



Which Age Groups Are Most Likely to Be Denied?

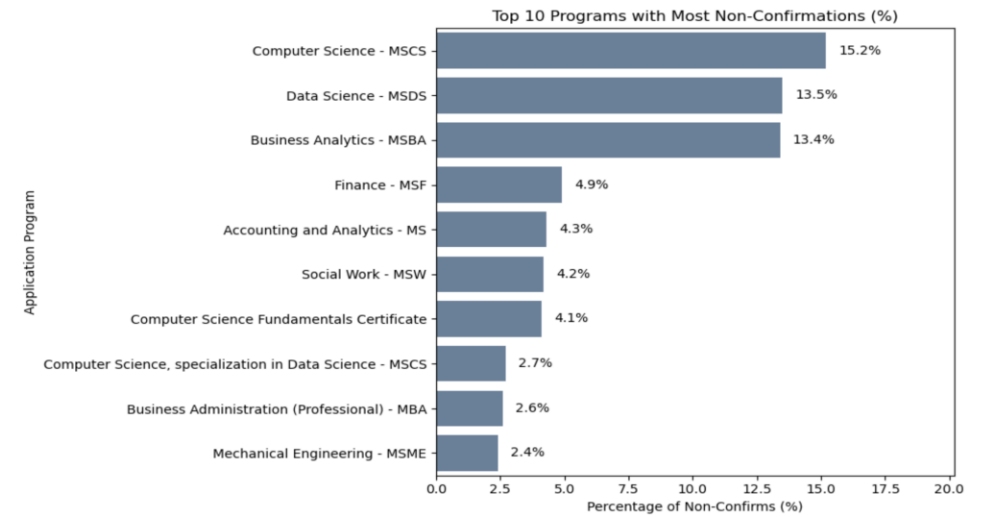
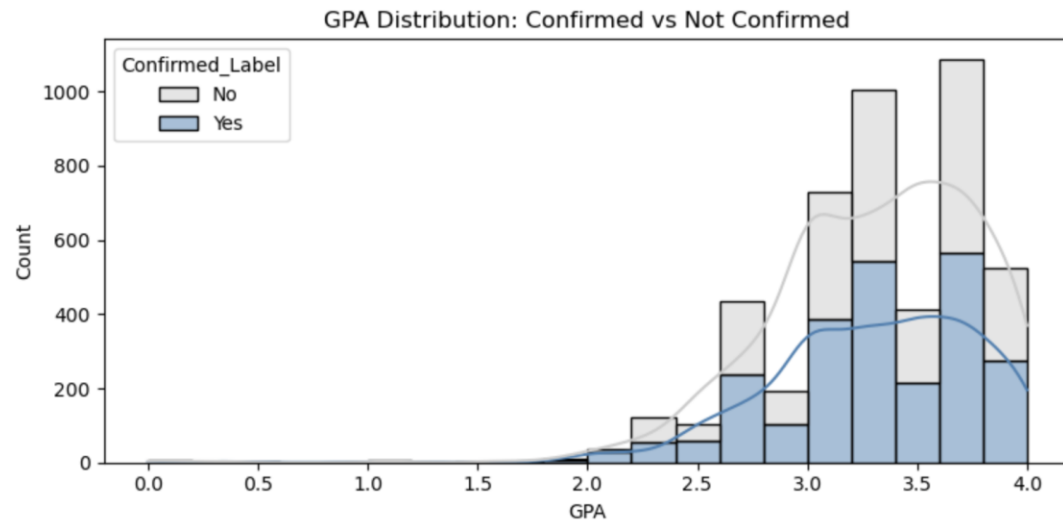
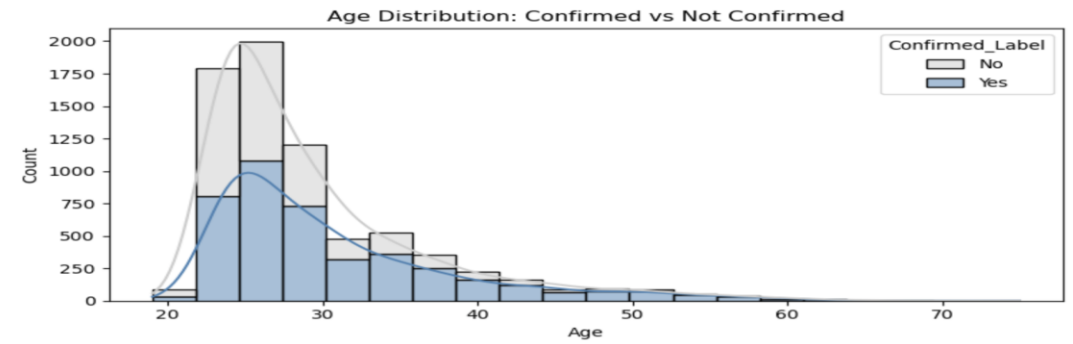
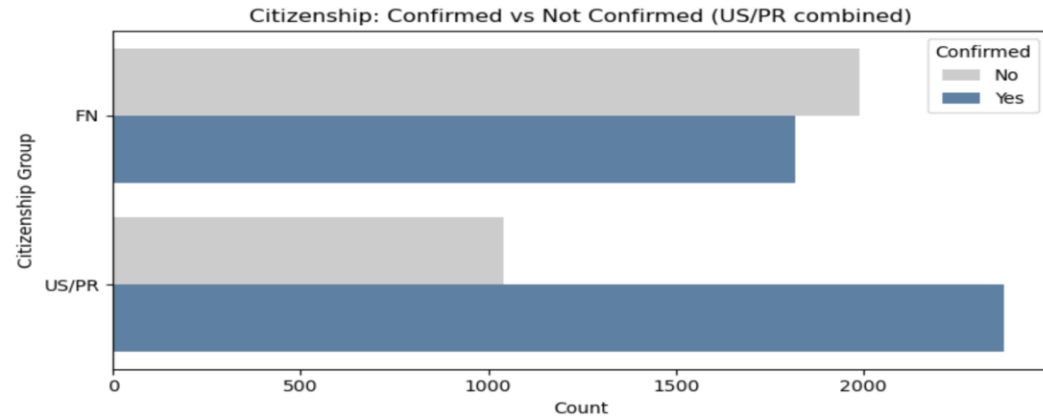
Denial rates are slightly higher among career changers (45+)



Who Gets In? Exploring GPA and Age Patterns in Graduate Admissions

Younger applicants cluster at higher GPAs, while denial patterns widen with age and lower GPA





Confirm but Don't Register

